

1. Student Details

Name: Mohd Zaid

Roll Number: [24f3001537](#)

Email: 24f3001537@ds.study.iitm.ac.in

About Me:

I am a student in the IIT Madras BS Degree program with a strong interest in web development and building real-world applications. I enjoy working with Flask, databases, and UI design to create systems that solve practical problems. Alongside this, I have a solid foundation in data structures and algorithms, which helps me write efficient, scalable, and optimized code. I am particularly interested in applying these concepts to backend development and system design.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

The objective of this project is to develop a web-based Placement Portal that connects students, companies, and administrators. The system allows companies to post job opportunities, students to apply, and administrators to manage and monitor the entire placement process efficiently.

Approach:

The application is developed using Flask as the backend framework along with SQLAlchemy ORM and SQLite database. The system is divided into modules for Student, Company, and Admin. Each role has its own functionalities such as registration, approval, job posting, and application tracking. The frontend is built using HTML, Jinja2 templates, and Bootstrap for styling.

3. AI / LLM Declaration

I used ChatGPT (GPT-5) to assist in:

- Debugging Flask routing and GitHub issues
- Improving UI design using Bootstrap
- Structuring database models and routes

The overall AI usage is approximately **20–25%**, mainly for guidance and debugging. All core logic, implementation, and integration were completed independently.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for database operations
SQLite	Database
Jinja2	Template rendering
Bootstrap 5	UI styling
HTML/CSS	Frontend design

5. Database Schema / ER Diagram

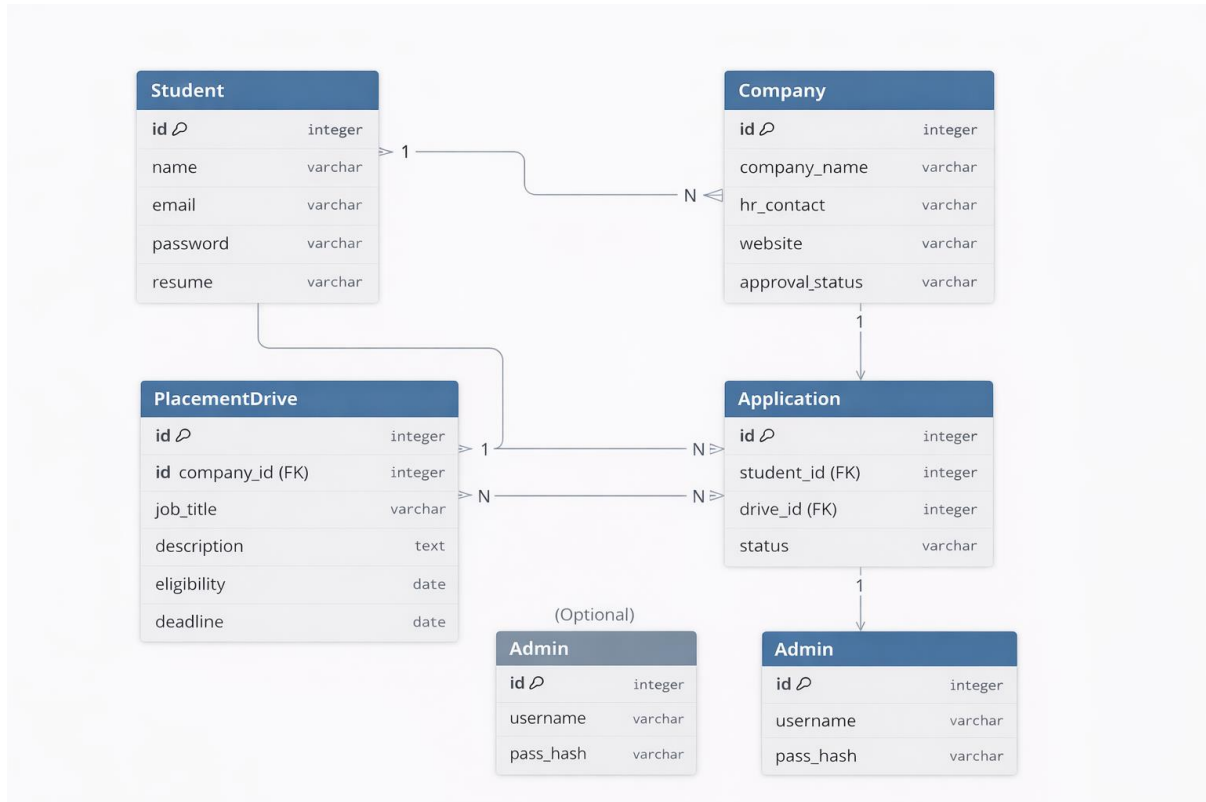
Tables:

- **Student**
(id, name, email, password, resume)
 - **Company**
(id, company_name, hr_contact, website, approval_status)
 - **PlacementDrive**
(id, job_title, description, eligibility, deadline, company_id)
 - **Application**
(id, student_id, drive_id, status)
-

Relationships:

- One-to-Many → Company → PlacementDrive
- One-to-Many → Student → Application
- One-to-Many → PlacementDrive → Application

ER diagram



6. Application Routes / Functional Endpoints

Route	Method	Description
/	GET	Home page
/student/register	GET/POST	Student registration
/company/register	GET/POST	Company registration
/admin/dashboard	GET	Admin dashboard
/admin/companies	GET	View companies
/admin/approve/<id>	GET	Approve company
/company/create_drive	GET/POST	Create placement drive

/drives	GET	View all drives
/apply/<id>	GET	Apply to drive
/applications	GET	View applications
/student/dashboard	GET	Student dashboard

7. Architecture and Features

Architecture Overview:

- `app.py` → Main Flask application
 - `models.py` → Database models
 - `templates/` → HTML pages
 - `static/` → CSS and assets
-

Implemented Features:

- ✓ Student registration
 - ✓ Company registration
 - ✓ Admin approval system
 - ✓ Placement drive creation
 - ✓ Job application system
 - ✓ Application tracking
 - ✓ Admin dashboard (counts of students, companies, drives)
-

Additional Improvements:

- ✓ Bootstrap-based UI
 - ✓ Navigation between pages
 - ✓ Background styling
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8. AI Usage Breakdown

Overall AI Usage: ~20–25%

Component	AI Usage %
Backend (Flask)	5%
Database (SQLAlchemy)	4%
Templates (HTML/Jinja)	4%
UI Styling (CSS/Bootstrap)	5%
Debugging	5–7%
GitHub Setup	3–5%

9. Video Presentation

Drive Link:

https://drive.google.com/file/d/1PshRweBOT7J3H2CLYBKnM1a9jD6VNBjH/view?usp=drive_link

10. Conclusion

This project demonstrates the implementation of a full-stack web application using Flask and SQLite. It successfully integrates multiple roles (Admin, Company, Student) and provides a structured system for managing placement activities. The project helped in understanding backend development, database design, and frontend integration.